Functional Annotation Report: paaB (Q88HS6 / PP_3277) in *Pseudomonas putida* KT2440

Gene / Protein Identity Verification

- **UniProt accession:** Q88HS6
- **Gene name / locus:** *paaB* / PP_3277
- **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950)
- **UniProt description:** "Ring 1,2-phenylacetyl-CoA epoxidase regulatory subunit," EC 1.14.13.149
- **Domains:** PaaB (Pfam PF06243; InterPro IPR009359, IPR038693)

Verification outcome — CONFIRMED. The gene symbol *paaB*, the EC number 1.14.13.149 (1,2-phenylacetyl-CoA epoxidase), the PaaB Pfam/InterPro domain, and the position of PP_3277 within the *P. putida* KT2440 *paa* gene cluster all consistently identify this protein as a subunit of the **phenylacetyl-CoA ring-1,2-epoxidase (monooxygenase) complex** of the aerobic phenylacetate degradation pathway. This is the correct gene; the literature cited below is specific to this protein family and pathway, which is explicitly documented in *P. putida* (Teufel et al. 2010, P 20660314).

A minor nomenclature caveat is noted in Limitations (subunit-letter usage differs slightly between organisms/authors).

1. Summary (Answer to the Research Question)

paaB encodes a small **structural/assembly subunit** of the multicomponent **phenylacetyl-CoA epoxidase (monooxygenase) complex, PaaABCE**, which catalyzes the committed, oxygen-dependent step of the bacterial aerobic phenylacetate catabolic pathway: insertion of oxygen into the aromatic ring of phenylacetyl-CoA to form ring-1,2-epoxyphenylacetyl-CoA (EC 1.14.13.149). PaaB is **not catalytic** — it carries no metal or substrate site — but instead forms a **homodimer that bridges two catalytic PaaAC heterodimers**, organizing the complex into an active heterohexamer. The complex operates in the **cytoplasm** on the CoA-thioester substrate as the ring-activating funnel step feeding phenylalanine, styrene, and environmental aromatics into central metabolism (acetyl-CoA + succinyl-CoA).

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2. Primary Function

2.1 The reaction and the complex

The phenylacetate (PA) pathway processes intermediates as **coenzyme A thioesters**. Free PA is first activated to **phenylacetyl-CoA (PA-CoA)** by a PA-CoA ligase (PaaK/PaaF). The central enzyme, the **four-subunit phenylacetyl-CoA monooxygenase (PA-CoA MO)**, then epoxidizes the aromatic ring:

> phenylacetyl-CoA + O_2 + NADPH → ring-1,2-epoxyphenylacetyl-CoA + $NADP^+$ + H_2O (EC 1.14.13.149)

The aromatic ring "becomes activated to a ring 1,2-epoxide by a distinct multicomponent oxygenase" (Teufel et al. 2010, P 20660314). Remarkably, the same complex is **bifunctional**: it can also run in reverse, mediating "the NADPH-dependent removal of the epoxide oxygen, regenerating phenylacetyl-CoA with formation of water," an intrinsic detoxification/escape mechanism to prevent accumulation of the reactive epoxide (Teufel, Friedrich & Fuchs 2012, P 22398448). Catalysis depends on a **carboxylate-bridged di-iron centre** housed in the oxygenase subunit PaaA.

2.2 Subunit roles and PaaB's specific function

The complex is a **new family of monooxygenases** that "differs in subunit organization from other monooxygenases" (Grishin et al. 2013, P 24055609).

Assigning the subunits:

- **PaaA** — the **oxygenase (catalytic) subunit**, containing the di-iron active site where ring epoxidation occurs.
- **PaaC** — a **structural subunit** that partners PaaA to form the catalytic **PaaAC heterodimer**.
- **PaaE** — the **NAD(P)H-dependent reductase** that supplies electrons to the di-iron centre; a conserved "lysine bridge" constellation (Lys68, Glu49, Glu72, Asp126) near the Fe site in PaaA forms part of the electron-transfer path from PaaE. Mutating this bridge reduced activity 20–50-fold (Grishin et al. 2013, P 24055609).
- **PaaB (this protein, Q88HS6)** — a **structural/scaffolding subunit**. Structural studies (low-resolution crystallography, electron microscopy, small-angle X-ray scattering) showed "the PaaACB complex forms **heterohexamers, with a homodimer of PaaB bridging two PaaAC heterodimers**" (Grishin et al. 2013, P 24055609).
  PaaB "is **unrelated to the small subunits of homologous monooxygenases**," making it a distinctive feature of this enzyme family.

Thus PaaB's precise role is **architectural/regulatory**: it does not bind iron, O_2, NAD(P)H, or the CoA substrate itself, but is required for correct **oligomerization and assembly** of the active PaaABCE complex. The UniProt label "regulatory subunit" reflects this non-catalytic, assembly/organizing function.

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3. Substrate Specificity

The complex acts specifically on **phenylacetyl-CoA (the CoA thioester), not on free phenylacetate**. The CoA-thioester activation both delivers the substrate to the enzymes and, mechanistically, is part of a "hybrid" strategy combining aerobic ring oxygenation with anaerobic-style CoA derivatization (Grishin & Cygler 2015, 26075354).

The PA pathway is the convergence point ("phenylacetyl-CoA catabolon") for many aromatic sources — phenylalanine, 2-phenylethylamine, styrene, lignin-related compounds, and man-made pollutants — all of which funnel into PA-CoA before the PaaABCE epoxidation step (Teufel et al. 2010, 20660314; Alonso et al. 2003, 14597173). PaaB, as a structural subunit, does not independently confer substrate specificity; specificity resides in the PaaA catalytic site.

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4. Subcellular Localization

The PaaABCE complex is a **soluble, cytoplasmic** enzyme. Its substrates and products are intracellular CoA thioesters, and all downstream steps (isomerization to an oxepin, hydrolytic ring opening, and β-oxidation-like chemistry) are carried out by soluble cytoplasmic enzymes (Teufel et al. 2010, 20660314). There is no evidence of membrane association or secretion; the complex has been purified and characterized biochemically and structurally as a soluble multiprotein assembly (Grishin et al. 2013, 24055609). Therefore PaaB carries out its scaffolding function **inside the cell, in the cytoplasm**.

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5. Pathway Context

The full aerobic PA / phenylacetyl-CoA catabolon operates as follows (Teufel et al. 2010, P 20660314; Grishin & Cygler 2015, P 26075354):

1. **Activation:** phenylacetate → phenylacetyl-CoA (PA-CoA ligase, PaaK/PaaF).
2. **Ring epoxidation (PaaB's step):** PA-CoA → ring-1,2-epoxyphenylacetyl-CoA by **PaaABCE** (di-iron monooxygenase; PaaB = structural subunit). The complex can also deoxygenate the epoxide back to PA-CoA to limit toxicity (Teufel et al. 2012, P 22398448).
3. **Isomerization:** epoxide → oxepin (a seven-membered O-heterocyclic enol ether) by a ring-1,2-epoxyphenylacetyl-CoA isomerase (PaaG).
4. **Hydrolytic ring opening** (PaaZ / oxepin-CoA hydrolase-aldehyde dehydrogenase).
5. **β-oxidation-like degradation** (PaaFGHIJ) → **acetyl-CoA** + **succinyl-CoA**, entering the TCA cycle.
6. **Detoxification support:** thioesterases **PaaI and PaaY** hydrolyze reactive intermediates to non-reactive products (Teufel et al. 2012, P 22398448).

This pathway is present in ~16% of sequenced bacterial genomes, including *E. coli* and *P. putida*, and is important both environmentally (styrene/aromatic pollutant remediation) and in the biology of certain pathogens where reactive early intermediates can contribute to virulence (Teufel et al. 2010, P 20660314).

In *Pseudomonas*, the *paa* cluster also connects to biotechnologically relevant processes such as polyhydroxyalkanoate (aromatic bioplastic) metabolism (García et al. 1999, P 10506180).

6. Evidence Base

- **Experimental (structural):** Crystallography + EM + SAXS defined the PaaACB heterohexamer and PaaB's bridging homodimer role; mutagenesis of the PaaA "lysine bridge" quantified electron-transfer importance (Grishin et al. 2013, P 24055609). Precise, low-throughput structural/biochemical evidence.
- **Experimental (biochemical/mechanistic):** Purification and characterization of PaaABCE demonstrated bifunctional oxygenase/deoxygenase activity and the di-iron centre (Teufel et al. 2012, P 22398448).
- **Experimental (pathway elucidation):** Isotope-labeling and intermediate identification established the epoxide→oxepin route across *E. coli*/*P. putida* (Teufel et al. 2010, P 20660314).
- **Genetic:** *paa* gene-cluster characterization and complementation in *Pseudomonas* strains (Alonso et al. 2003, P 14597173; Ferrández et al. 1998, P 9748275; Mohamed et al. 2002, P 12189419).
- **Bioinformatic/evolutionary:** PaaB defines a distinct InterPro/Pfam family (PF06243) unrelated to small subunits of other monooxygenases, underscoring its unique structural role (Grishin et al. 2013, P 24055609).

7. Supported and Refuted Hypotheses

Supported - PaaB is a subunit of the PaaABCE phenylacetyl-CoA ring-1,2-epoxidase (EC 1.14.13.149). ✔ - PaaB is a **structural/assembly (non-catalytic) subunit** forming a homodimer bridging two PaaAC catalytic dimers. ✔ - The complex functions in the **cytoplasm** on the **CoA-thioester** substrate. ✔ - The complex is bifunctional (epoxidase + deoxygenase) for epoxide detoxification. ✔

Refuted / Not supported - PaaB is the catalytic (metal-binding) or substrate-binding subunit — **refuted**; catalysis resides in PaaA's di-iron site. ✘ - PaaB acts on free phenylacetate — **refuted**; the substrate is phenylacetyl-CoA. ✘

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8. Limitations and Future Directions

- **Nomenclature:** Subunit letters differ between organisms and authors (e.g., "PaaABCDE" in *E. coli*/*Azoarcus* vs. "PaaACBE" in Grishin's structural work). This report maps the *P. putida* PP_3277 *paaB* gene (Q88HS6) to the structurally characterized bridging subunit; direct crystallographic work was performed largely on orthologous complexes, so the assignment relies on strong sequence/domain conservation across the family rather than a *P. putida*-specific structure.
- **Direct localization data** (e.g., fractionation) for *P. putida* PaaB specifically are inferred from the soluble nature of the purified complex; explicit *in vivo* localization studies would strengthen this.
- **Mechanistic role of PaaB in regulation vs. pure assembly** is not fully resolved — whether PaaB modulates the oxygenase/deoxygenase balance beyond simply enabling oligomerization is an open question worth targeted biochemical study.

References

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